

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE:CSA1603

COURSE NAME: Data Warehousing and Data Mining

NAME: P. GAYATHRI

REG.NO: 192425277

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total(10)
<i>Weightage</i>	20%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I __P.GAYATHRI/192425277 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.GAYATHRI

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)

Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

ANSWERS:

1)

CSA1603 - Data Warehouse and Mining

CO1 - ATL-3

Names: P. Gayathri

Reg No: 192025237

- ① Compare k-means and hierarchical clustering using this data-set and explain.

K-means vs Hierarchical clustering

K-means clustering divides the customer into a fixed no. of clusters using centroids.

* It is fast and works well for large datasets.

* It repeatedly assigns customers to the nearest centroid and updates the centroid.

Hierarchical clustering

It creates clusters step-by-step.

⇒ It uses a dendrogram to show relationships between customers.

⇒ It is slower and requires more computation for large datasets.

Conclusion

K-means is better for large datasets customer data.

Large hierarchical clustering is useful for understanding cluster relationships.

2)

② Compare k-means and EM algorithm in handling overlapping clusters for the given dataset.

Q. K-Means vs EM Algorithms

K-means:

K-means assigns each document to only one cluster. It works better when clusters are clearly separated.

EM gives a probability of a document belonging to different clusters.

Therefore, a document containing word E_i from two topics can have membership in both clusters. This is useful for the given document data because some documents may have overlapping word patterns.

Conclusion:

EM is better than k-means for handling overlapping documents clusters.

3)

compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results.

Ans Geographic Data - Euclidean vs Manhattan

Given:

P₁ - P₅ have latitude values from 12.97 - 13.10 and longitude values from 80.20 - 80.30

* Euclidean distance measures the direct straight line distance between two geographic points.

* Formula:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

* Manhattan distance adds the absolute latitude and longitude difference

* Formula

$$d = |x_1 - x_2| + |y_1 - y_2|$$

Conclusion

Euclidean distance is more suitable here because the data represents geographic location and direct spatial closeness is important.

4)

compare agglomerative and divisive clustering approaches and explain their suitability for this dataset.

Retail Products - Agglomerative vs Divisive

Given sales = 10,000, 25,000 and frequency

- * Agglomerative clustering starts with each product as a separate cluster.
- * It gradually merges similar products based on sales and frequency.
- * For example, $P_2 (£50,000, 22)$ and $P_4 (£52,000, 12)$ have high sales and can be considered relatively similar.
- * Divisive clustering starts with all five products in one cluster.
- * It then splits the products into smaller groups.
- * Agglomerative clustering is simple and suitable for grouping retail products based on sales and frequency.
- * For example $P_2 (£50,000, 22)$
 $P_4 (£52,000, 12)$
have high sales and can be considered as similar.

vivo T3x
Aug 22, 2025

5)

⑤ Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types.

Mixed Attributes - Centroid-based vs Density Based.

Given: Price = ₹ 220 - ₹ 550

Category = A/B.

Rating = 3.5 - 4.7

- * The dataset contains numerical attributes such as Price and Rating.
- * It also contains a categorical attributes, category (A/B).
- * Centroid-based clustering, such as K-means, forms groups around a central point.
- * K-means mainly works with numerical values, so category must be properly encoded.
- * Density-based on dense groups of products.

Conclusion:

Centroid-based clustering is simple for numerical attributes while density-based clustering is useful when the products form irregular groups or contain outliers.